

Jade Marie Ducharme

Curriculum Vitae

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Education

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| 2022– | Brown University
<i>Ph.D. (Physics)</i>
Advisor: Prof. Jonathan C. Pober |
| 2025 | Brown University
<i>M.Sc. (Physics)</i> |
| 2022 | McGill University
<i>B.Sc. (Physics)</i> |

Research Interests

- Radio interferometry, 21-cm cosmology, and the Epoch of Reionization
- Radio frequency interference mitigation techniques
- Machine learning for astronomy
- Bayesian modeling and probabilistic programming

Publications

Throughout the following sections, my name is indicated in **bold**.

- (2) Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Jade M. Ducharme, Jonathan C. Pober, and Michael J. Wilensky.
Submitted to PASA
- (1) Altitude Estimation of Radio Frequency Interference Sources via Interferometric Near-Field Corrections
Jade M. Ducharme & Jonathan C. Pober.
2025, *PASA*, 42, e010. DOI: <https://doi.org/10.1017/pasa.2024.123>.

Collaboration Publications

The following are papers in which my authorship results from participation in a collaboration.

- (3) The Murchison Widefield Array Phase III upgrade: Sensitivity doubled, number of baselines quadrupled, flexibility enhanced, and EoR observations optimised
S.J. Tingay, M. Johnston-Hollitt, R.B. Wayth, T.A. Booler, J. Jones, Y. Wu, J. Gan, G. Sleap, A. McPhail, C. Wintle, A. Williams, C.J. Phillips, L. Verduyn, D. Emrich, P. Giersch, C.J. Riseley, S. Duchesne, C.M. Trott, D. Null, B.W. Meyers, C.D. Nunhokee, N. Barry, L. Dressler, **J. Ducharme**, B. Hazelton, M. Lee, E. Lilleskov, M. Morales, J. Pober, Z. Shen, X. Wu, X. Hong, M.D. Filipović, S.E. Tremblay, M. Walker.
2026, *PASA*, 43, e085. DOI: <https://doi.org/10.1017/pasa.2026.10218>.
- (2) Improved Limits on the 21 cm Signal at $z = 6.5\text{--}7.0$ with the Murchison Widefield Array Using Gaussian Information
Cathryn M. Trott, C. D. Nunhokee, D. Null, N. Barry, Y. Qin, R. B. Wayth, J. L. B. Line, C. H. Jordan, B. Pindor, J. H. Cook, J. Bowman, A. Chokshi, **J. Ducharme**, K. Elder, Q. Guo, B. J. Hazelton, W. Hidayat, T. Ito, D. Jacobs, E. Jong, M. Kolopanis, T. Kunicki, E. Lilleskov, M. F. Morales, J. Pober, A. Selvaraj, R. Shi, K. Takahashi, S. J. Tingay, R. L. Webster, S. Yoshiura, and Q. Zheng.
2025, *ApJ*, 991, 211. DOI: <https://doi.org/10.3847/1538-4357/adff80>.
- (1) Limits on the 21 cm Power Spectrum at $z = 6.5\text{--}7.0$ from Murchison Widefield Array Observations
C. D. Nunhokee, D. Null, C. M Trott, N. Barry, Y. Qin, R. B. Wayth, J. L. B. Line, C. H. Jordan, B. Pindor, J. H. Cook, J. Bowman, A. Chokshi, **J. Ducharme**, K. Elder, Q. Guo, B. Hazelton, W. Hidayat, T. Ito, D. Jacobs, E. Jong, M. Kolopanis, T. Kunicki, E. Lilleskov, M. F. Morales, J. C. Pober, A. Selvaraj, R. Shi, K. Takahashi, S. J. Tingay, R. L. Webster, S. Yoshiura, and Q. Zheng.
2025, *ApJ*, 989, 57. DOI: <https://doi.org/10.3847/1538-4357/adda45>.

Conferences and Talks

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| July 2026 | Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Talk — MWA Project Meeting 
<i>Brown University, Providence, RI, USA.</i> |
| June 2026 | Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Poster — Canadian Astronomical Society's Annual General Meeting 
<i>University of Montréal, Montréal, QC, Canada.</i> |
| Feb. 2026 | Mitigating Radio Frequency Interference in Radio Astronomy: Recent Progress at Brown
Invited Talk — MIT Monday Afternoon Science Talks 
<i>MIT Kavli Institute, Cambridge, MA, USA.</i> |
| Jan. 2026 | Altitude estimation of radio frequency interference sources via interferometric near-field corrections
Talk — National Radio Science Meeting  |

Boulder, CO, USA.

Aug. 2024 U.S.-Based Research on Mitigating Radio Frequency Interference for MWA Data
Invited Talk — MWA Project Meeting [↗](#)
Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland.

Honors and Awards

2026 *Brown University*
Physics Merit Fellowship
Full funding for one academic semester

2025 *Brown University*
Forrest Award for Excellent Work Related to Experimental Apparatus
USD \$250

2024 *MWA Collaboration*
Awarded Best Talk at MWA Project Meeting

2023–2025 *National Science Foundation*
Collaborative Research: 21 cm Reionization Science with the MWA
Award #2106510 (full-year academic funding)

2021 *McGill Space Institute*
Summer Research Showcase Prize Winner
CAD \$250

2021 *Natural Sciences and Engineering Research Council of Canada*
Undergraduate Student Research Award
CAD \$7000

Professional Memberships

2023– *MWA Collaboration* [↗](#)
Collaborator

Teaching and Mentoring

Summer 2026 Co-supervised undergraduate student Neal Frankenberg
Brown University, Providence, RI, USA.

Spring 2026 Co-supervised undergraduate student Jacob Lipman
Brown University, Providence, RI, USA.

Summer 2025 Co-supervised undergraduate student Andrei Li
McGill University, Montréal, QC, Canada.

Spring 2023 Teaching Assistant – PHYS 0220 (Astronomy)

Brown University, Providence, RI, USA.

Fall 2022 Teaching Assistant – PHYS 0270 (Introduction to Astronomy)
Brown University, Providence, RI, USA.

Professional Development

2026 INTERSECT Research Software Engineering Bootcamp [↗](#)
Princeton University, NJ, USA

2026 Code/Astro Workshop [↗](#)
UC Santa Cruz, CA, USA

2024, 2025 AI Winter School [↗](#)
Brown University, Providence, RI, USA

2024 Precision Calibration Workshop [↗](#)
McGill University, Montreal, QC, Canada

Technical Skills

- **Programming:** Python, IDL, Stan, Bash, HTML/CSS, JavaScript
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn
- **High-Performance Computing:** SLURM workload management, distributed computing workflows
- **Radio Astronomy Software:** CASA, FHD/eppsiilon, pyuvdata, WSClean
- **Tools:** Git, Linux, React
- **Spoken languages:** French (native), English (fluent)

Public Outreach

2024– Event coordinator & lead web developer
Astronomy on Tap – Rhode Island Chapter (aot-ri.vercel.app).
Providence, RI, USA.